

Increasing Profitability

Airline Carriers



Stakeholders: Airline Carriers

Profitability

Revenue - Costs



DELTA

Alaska
AIRLINES

SINGAPORE
AIRLINES



UNITED



*How to increase
revenue?*

spirit



中国南方航空
CHINA SOUTHERN



RYANAIR

Emirates



Lufthansa

*How to decrease
costs?*

BRITISH AIRWAYS



AIRFRANCE



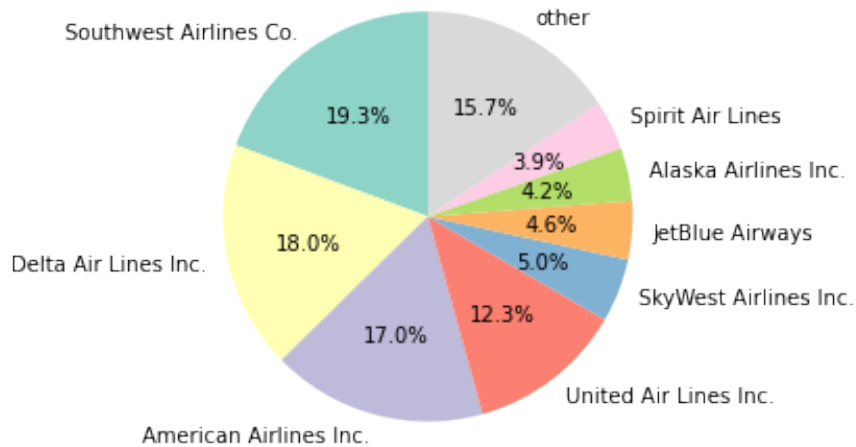
Southwest

American Airlines

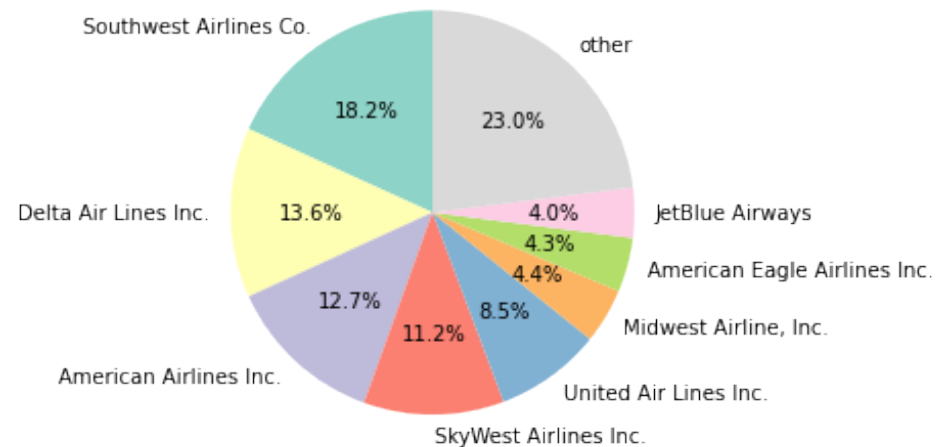


Market Share

Distribution of Passengers Among Different Airline Carriers



Distribution of Flights Among Different Airline Carriers



Data Set & Motivation

2019 Flights Delays - 6,489,062 rows (slimmed to 1,489,062)

Airports, Airlines, Plane Size and Weather

Strategy

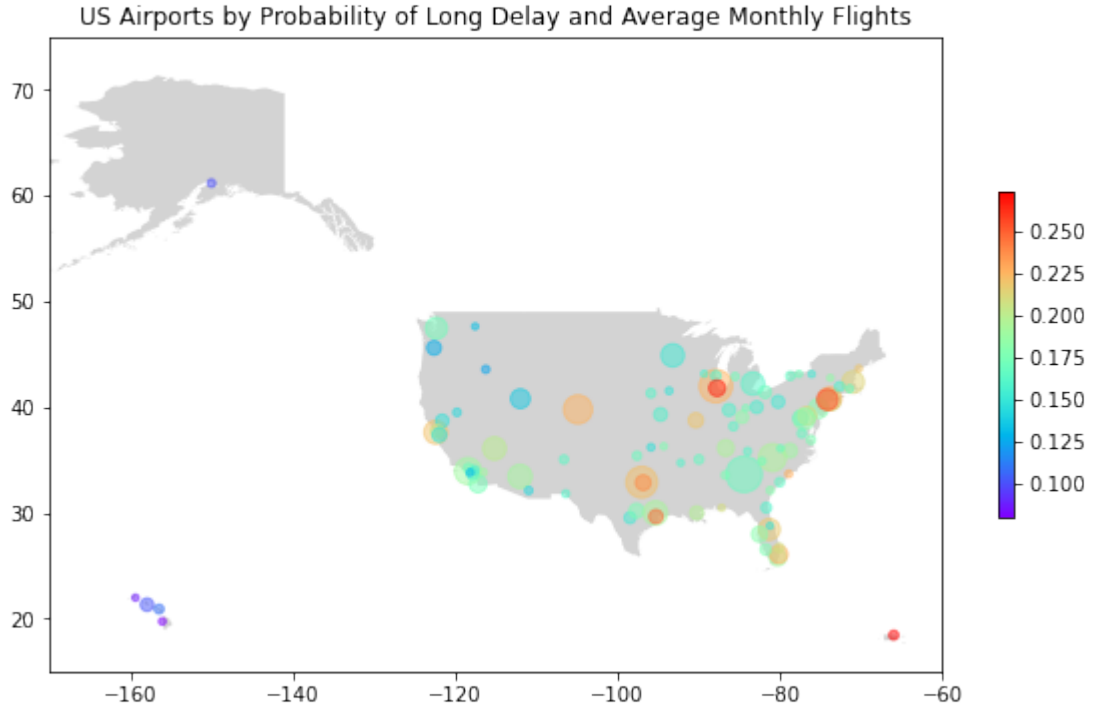
↓ Delays = ↑ Reputation, Loyalty, Market Share = ↑ Revenue

↓ Delays = ↓ Rescheduling, Vouchers = ↓ Costs

United States Airports

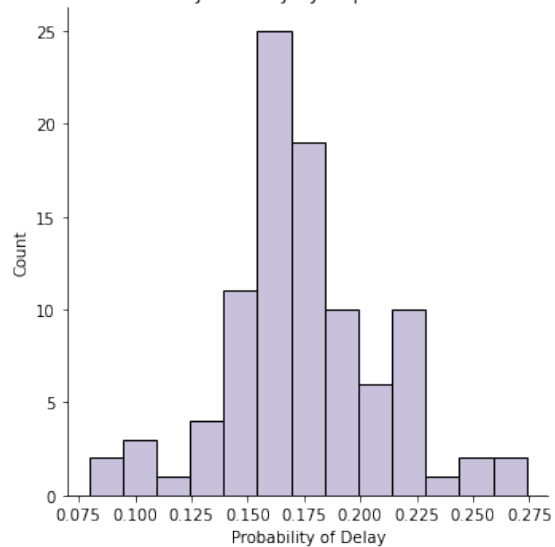
Exploratory Data Analysis

Do airports have different probabilities of delayed flights?

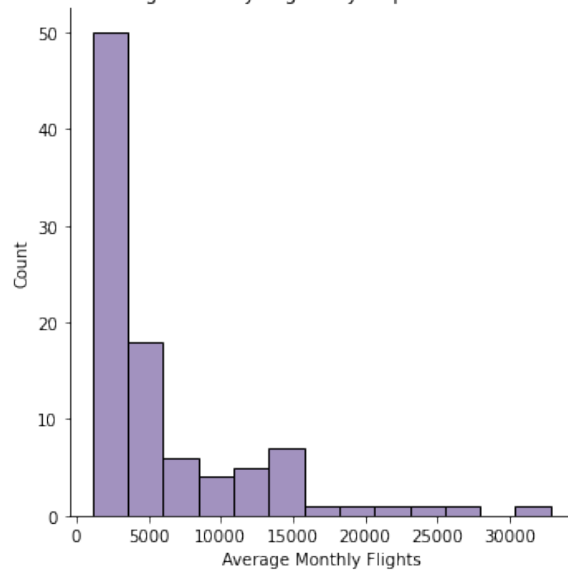


Distributions

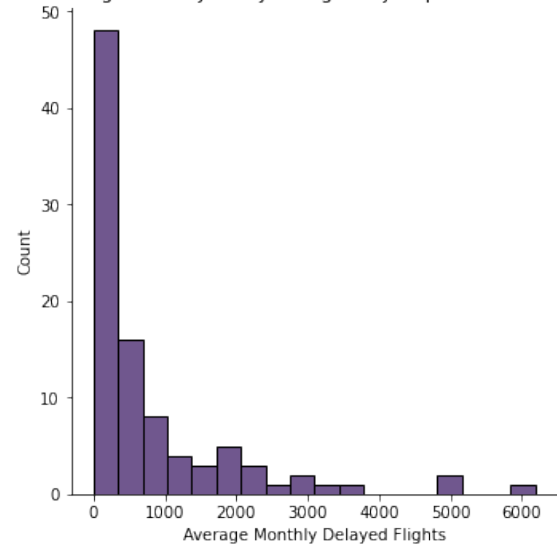
Probability of Delay by Airport Distribution



Average Monthly Flights by Airport Distribution



Average Monthly Delayed Flights by Airport Distribution



Regions

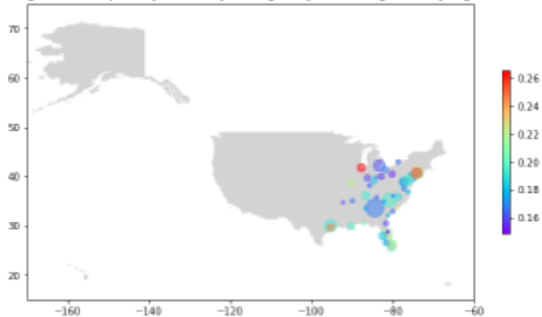
Region 1: US Airports by Probability of Long Delay and Average Monthly Flights



Region 3: US Airports by Probability of Long Delay and Average Monthly Flights



Region 2: US Airports by Probability of Long Delay and Average Monthly Flights



Region 5: US Airports by Probability of Long Delay and Average Monthly Flights



Region 4: US Airports by Probability of Long Delay and Average Monthly Flights



Region 6: US Airports by Probability of Long Delay and Average Monthly Flights



Region 7: US Airports by Probability of Long Delay and Average Monthly Flights

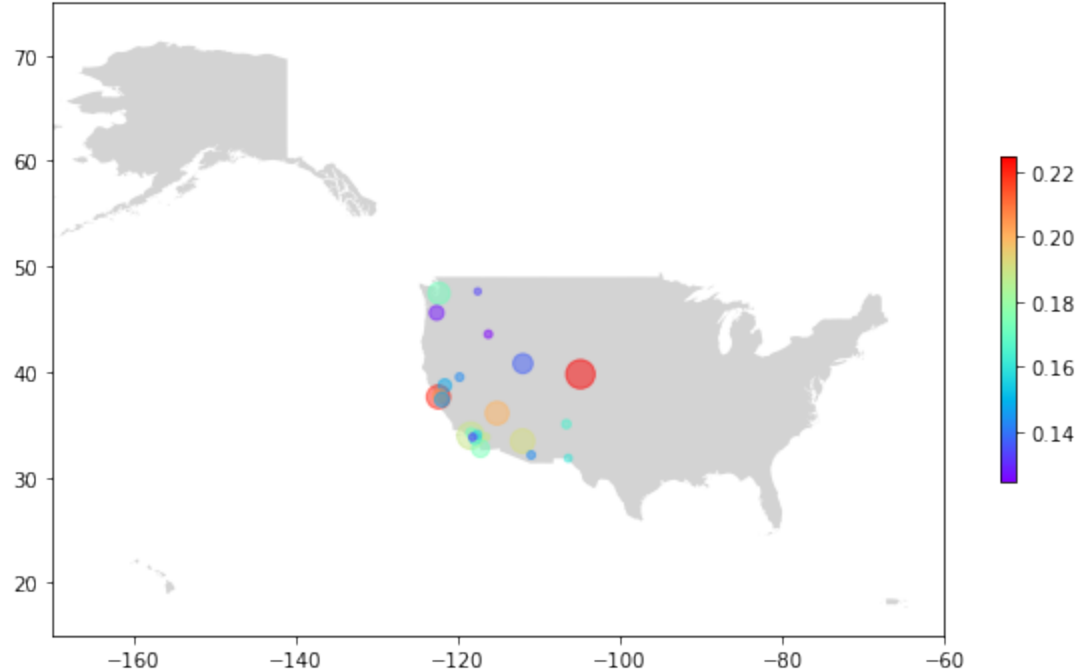


Best Region

Average probability of
delay:16.53%

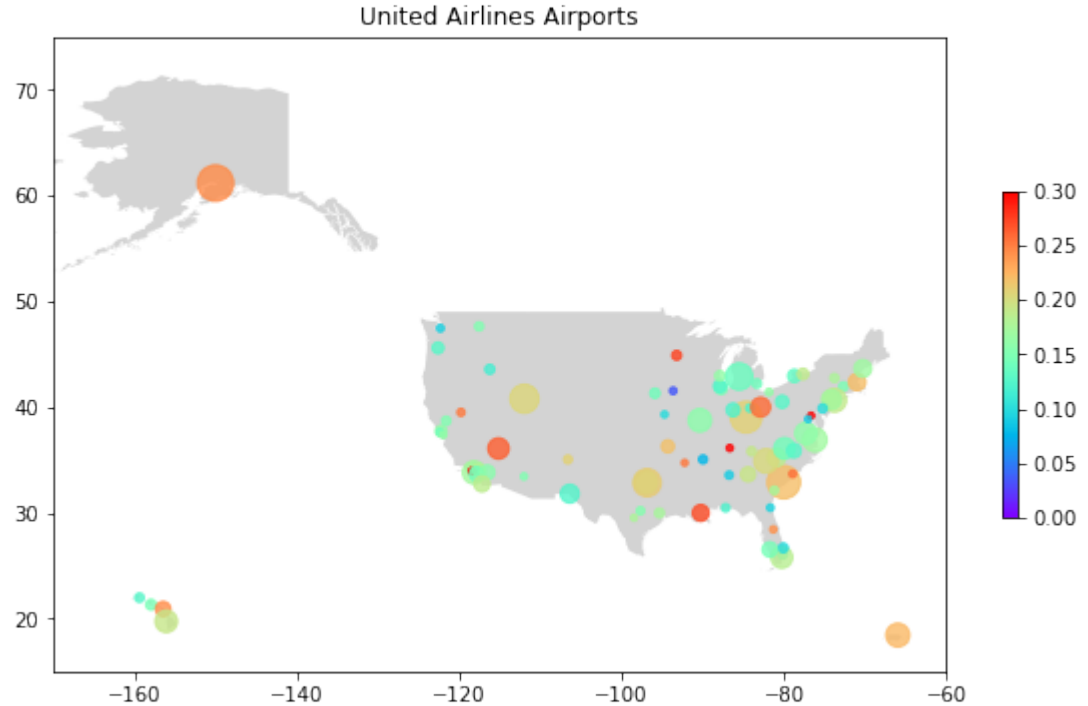
Average Monthly
Flights:6,483

Region 1: US Airports by Probability of Long Delay and Average Monthly Flights



Case Study

How to decrease the number of delayed flights for United Airlines by airport selection?



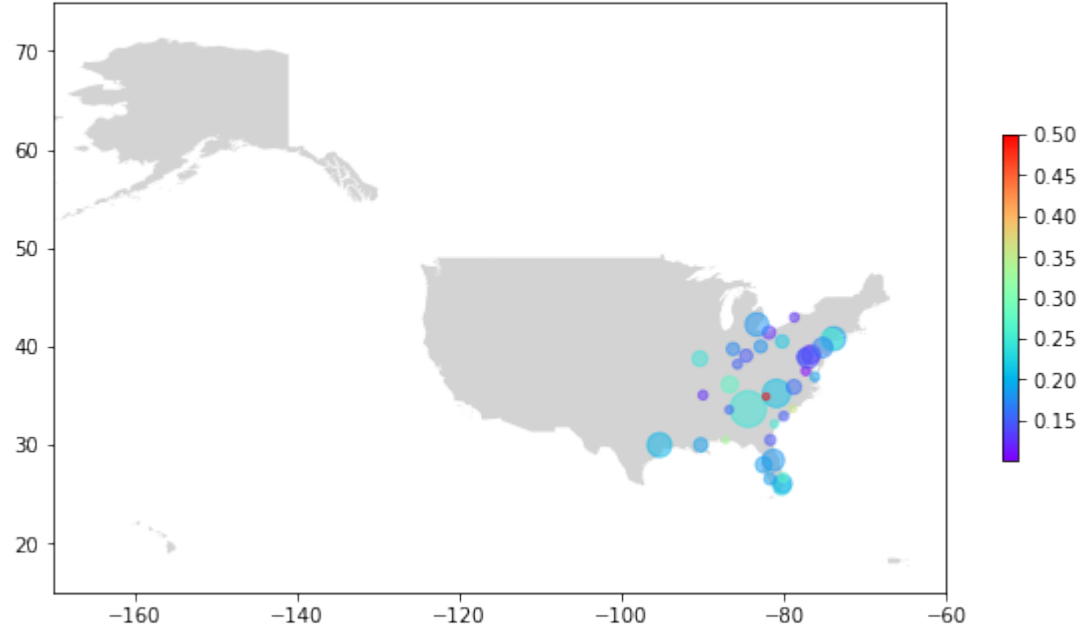
Worst Performing Region

Average Probability of delay:
20.0%

+1.4% Average

How to lower probability of delays?

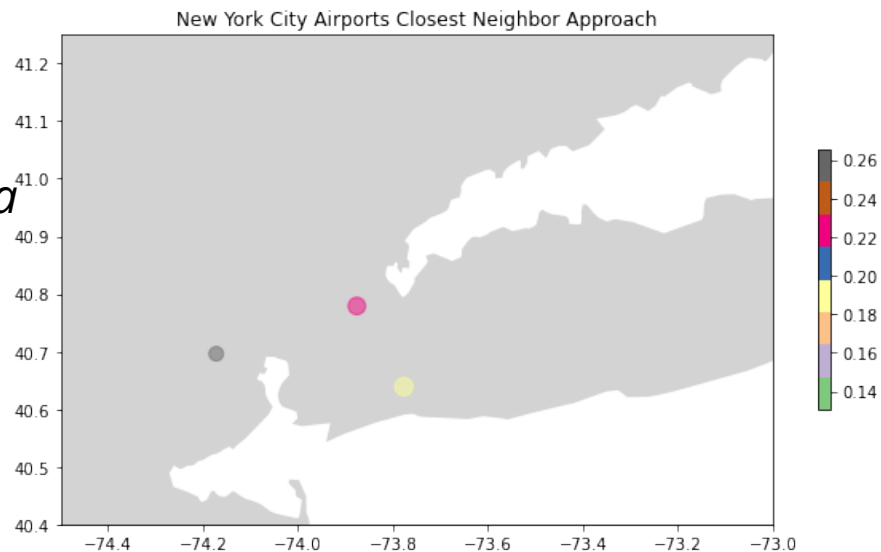
Region 2: US Airports by Probability of Long Delay and Average Monthly Flights



Closest Neighbor Approach

NYC area airports

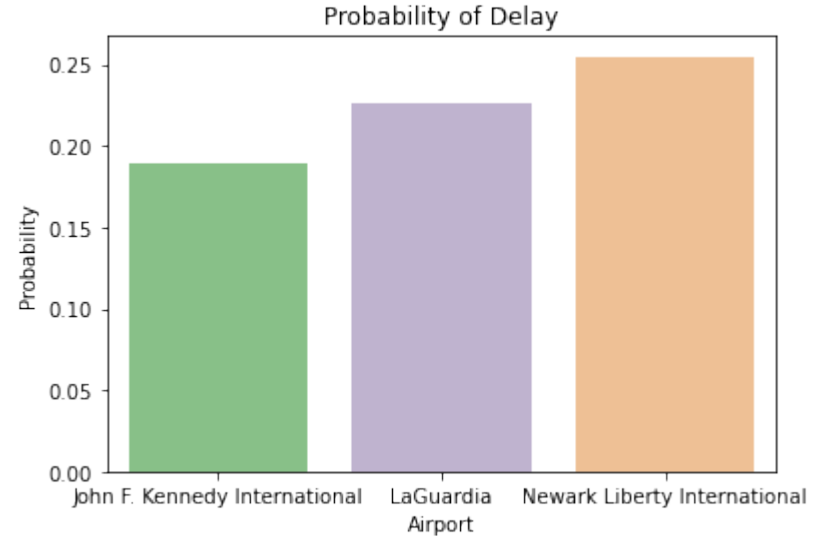
For flights with the same destination, funnel a higher proportion of flights to the airport with a lower probability of delay.



Differences Between Closest Neighbor

Consumers will attribute lower delays with the airline, not the airport

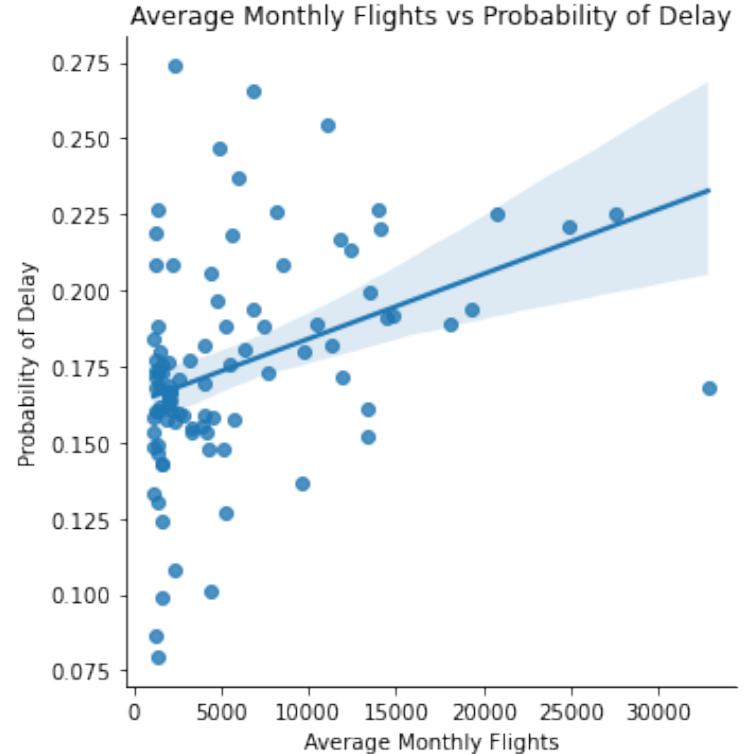
↑
Reputation, Loyalty, Revenue



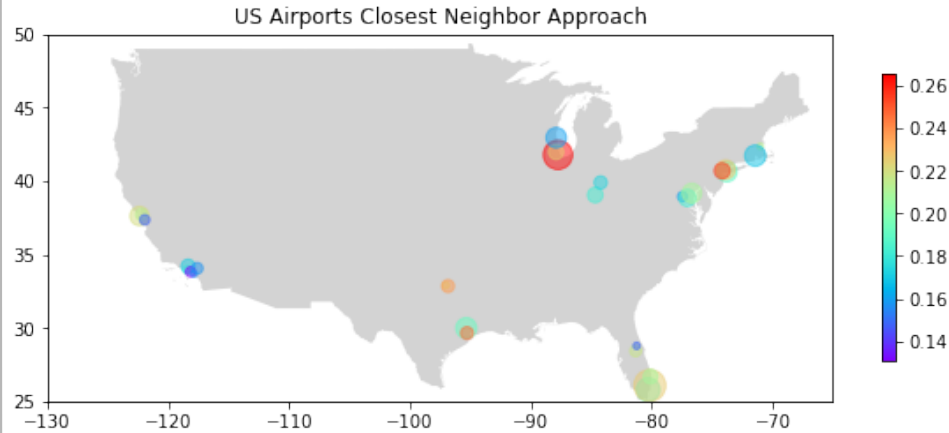
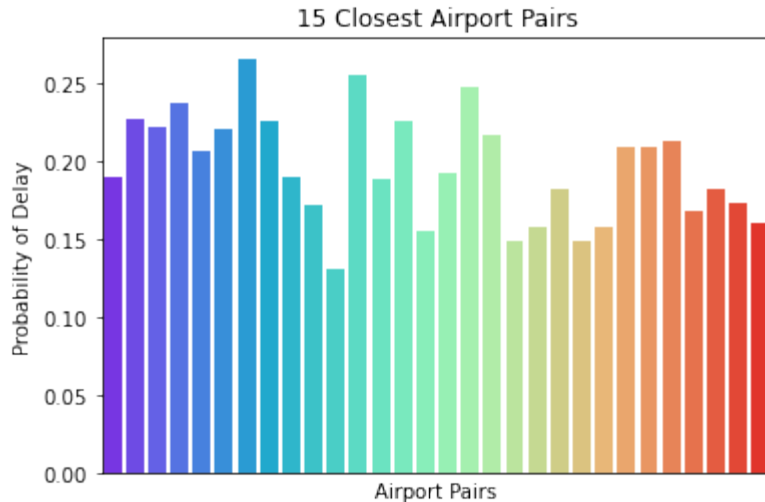
Correlation: Delays & Flights

Will increasing the number of flights of an airport increase the probability of delay?

Target smaller airports



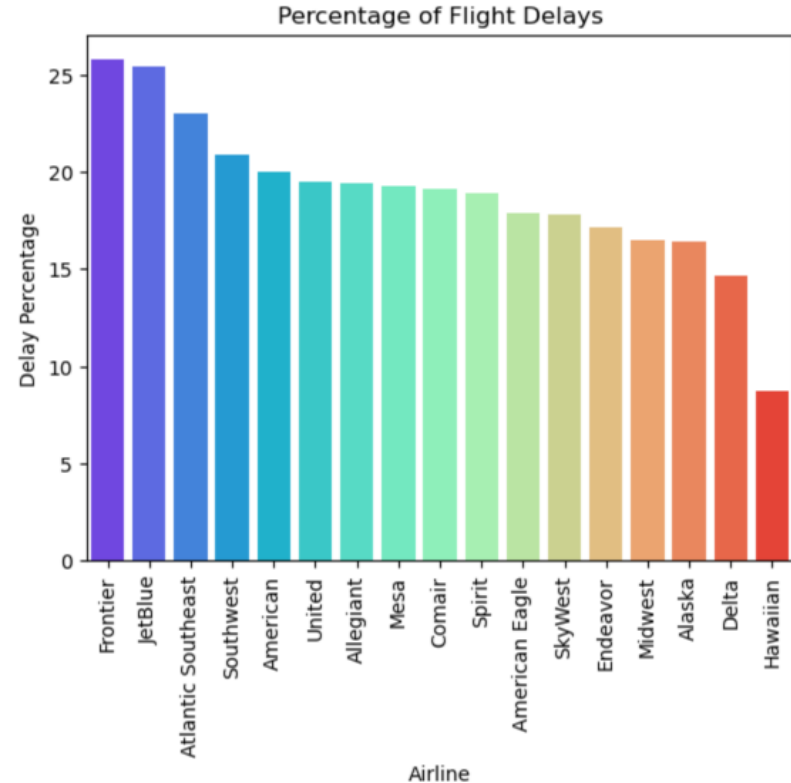
Closest Neighbor Approach-30 airports



United States Airlines

Exploratory Data Analysis

Which airlines have the most delays and what are their commonalities?



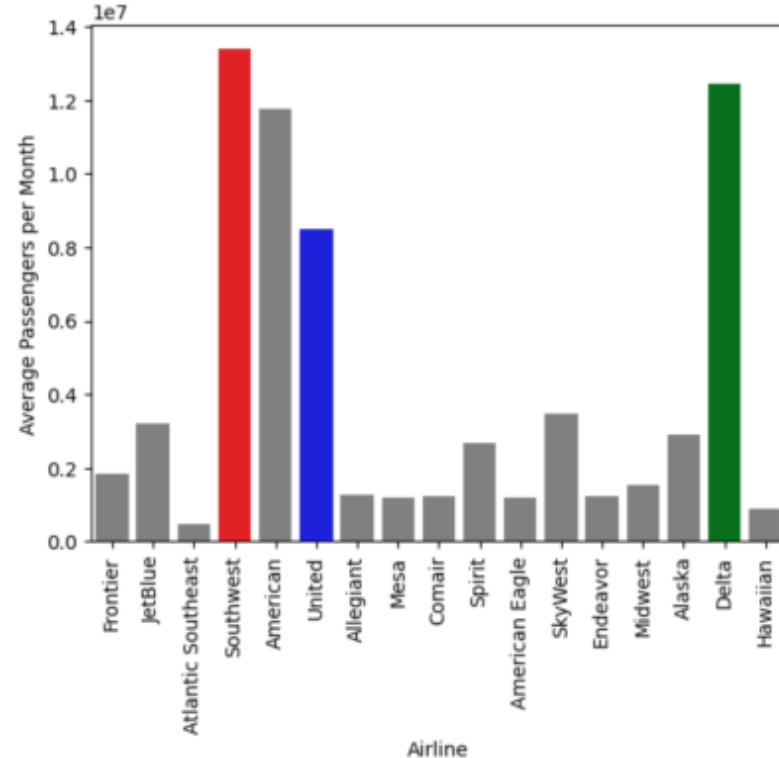
Case Study - Setting Benchmarks

Comparing profit generating variable,
passengers:

- United: 8,501,631
- Southwest: 13,382,999
- Delta: 12,460,183

Performance:

- -4,881,368 less than Southwest
- -3,958,552 less than Delta



Case Study - Delays

Delay Percentages:

Maximum: 25.75%

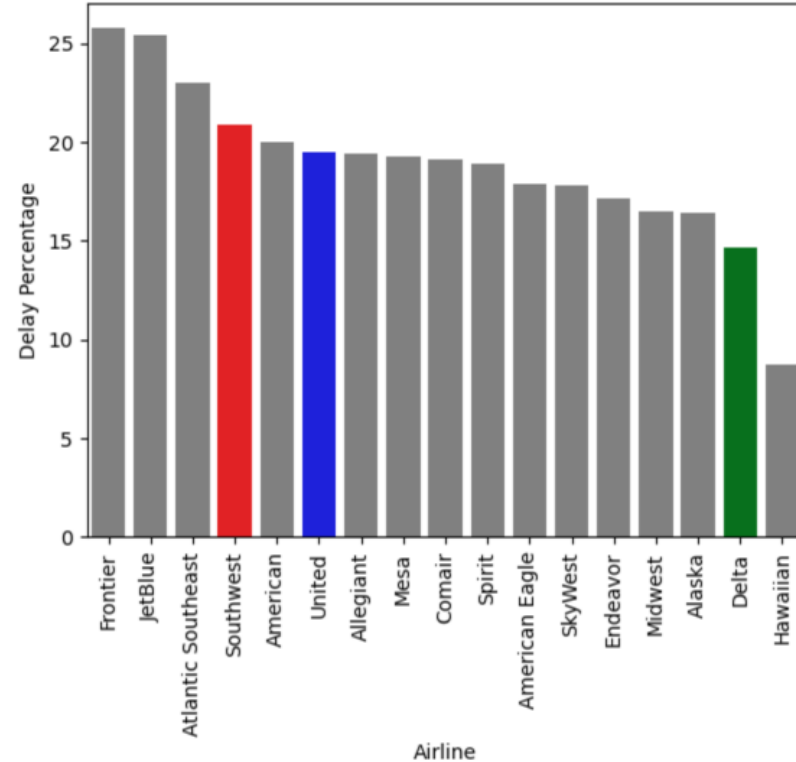
United: 19.46%

Southwest: 20.88%

Delta: 14.67%

Performance:

- -6.29% on upper bound
- +4.79% on lower bound



Case Study - Operation Variables

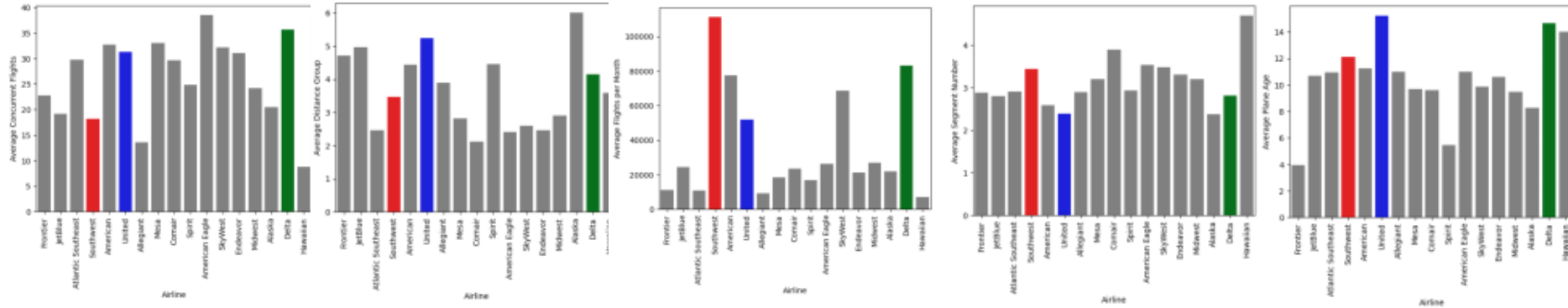
Concurrent Flights

Distance Group

Flights Per Month

Segment Number

Plane Age



Case Study - Operation Variable of Interest

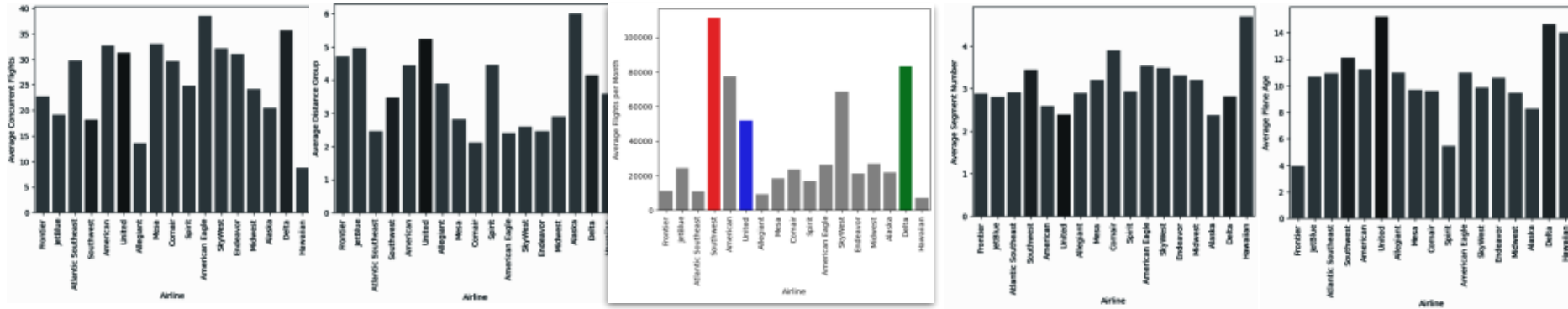
Concurrent Flights

Distance Group

Flights Per Month

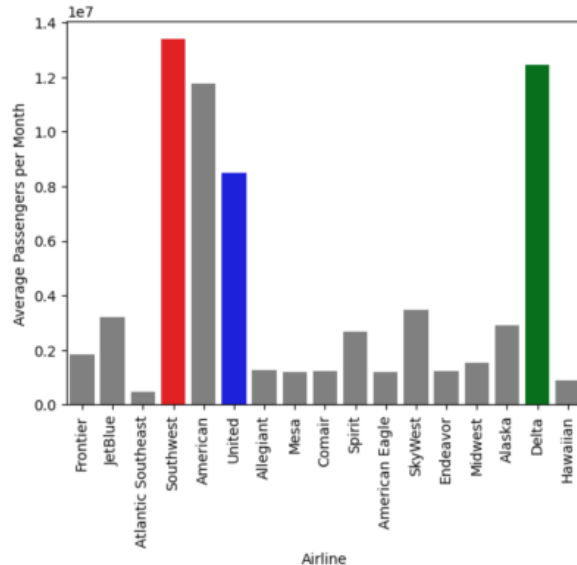
Segment Number

Plane Age

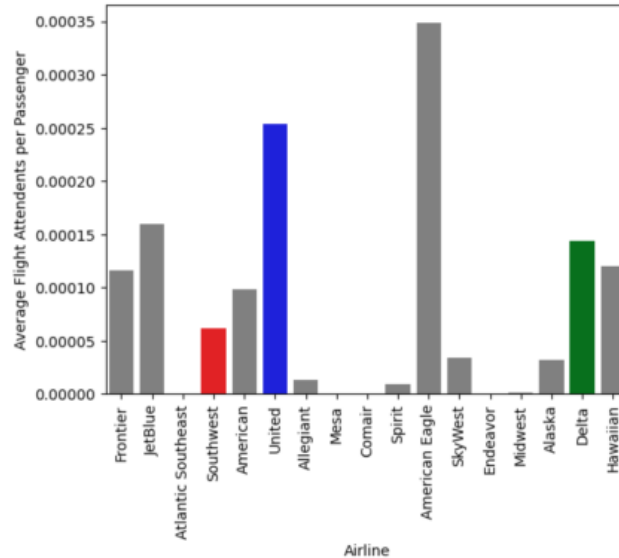


Case Study - Human Resources Variables

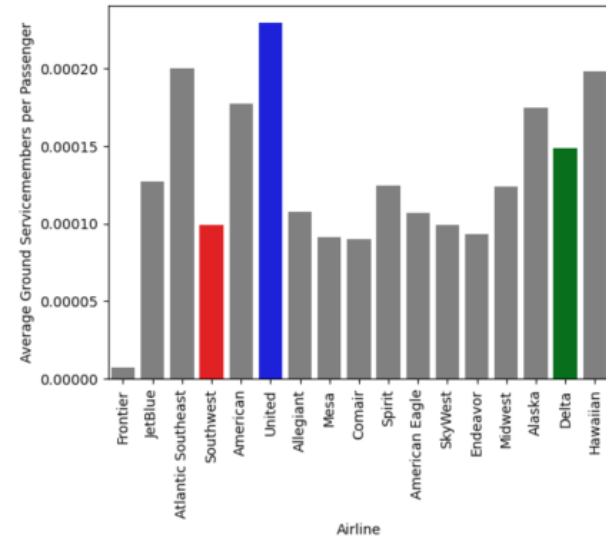
Passengers per Month



Flight Attendants per Passenger

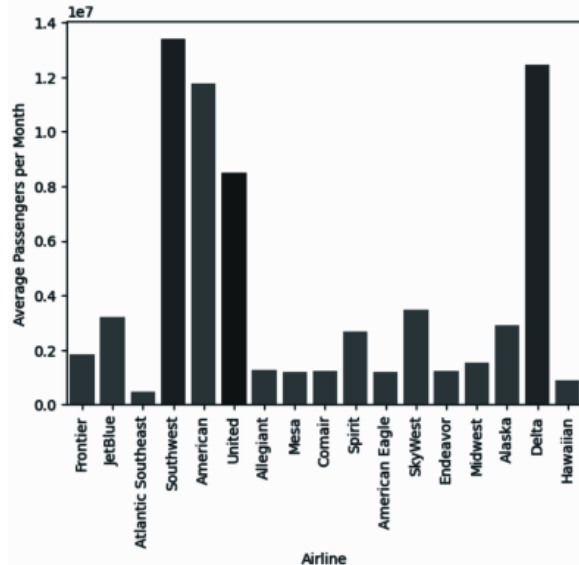


Ground Crew per Passenger

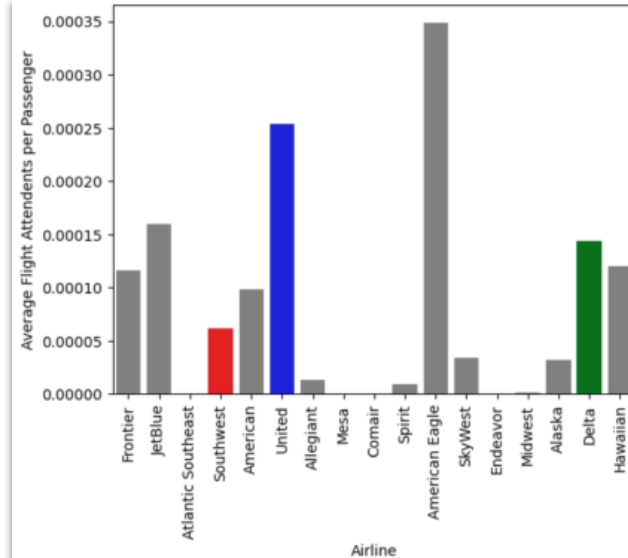


Case Study - Human Resources VOI

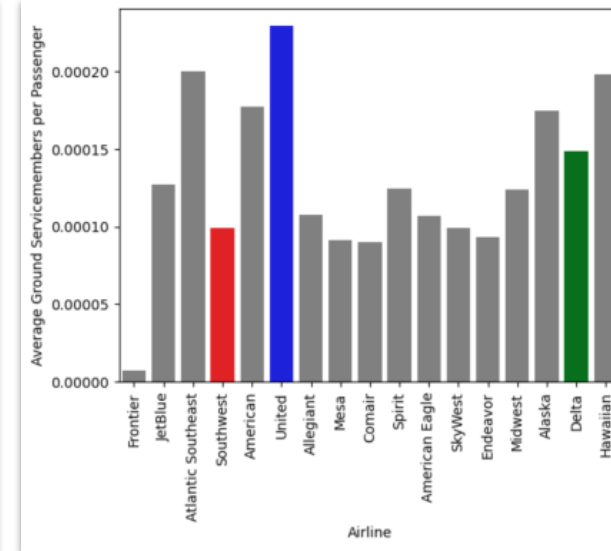
Passengers per Month



Flight Attendants per Passenger



Ground Crew per Passenger



Case Study - Flight Attendants per Passenger

United: $\sim 2.54 \times 10^{-4}$

Southwest: $\sim 6.2 \times 10^{-5}$

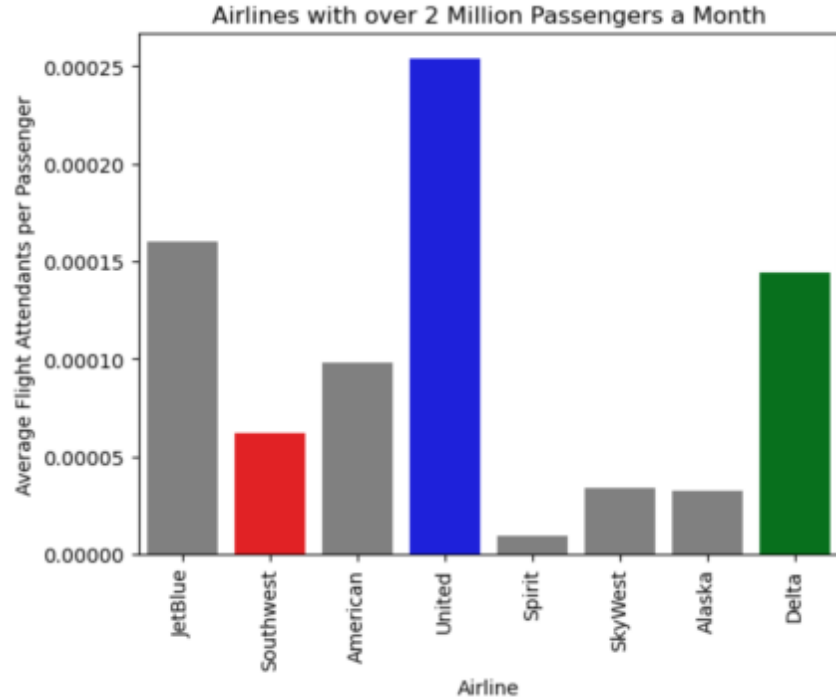
Delta: $\sim 1.44 \times 10^{-4}$

Average: $\sim 8.2 \times 10^{-5}$

Average, similar airlines: $\sim 9.9 \times 10^{-5}$

United is employing far more flight attendants than all of its competitors, with no discernible difference in delay time

Recommendation: United should slim down its flight attendant crews



Case Study - Ground Crew per Passenger

United: $2.29\text{e-}4$

Southwest: $\sim 9.9\text{e-}5$

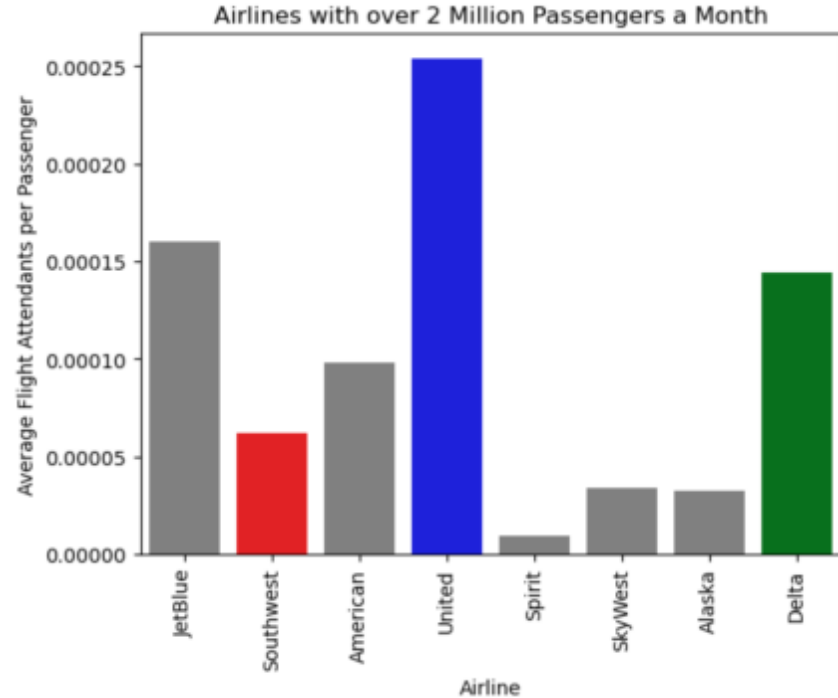
Delta: $1.49\text{e-}4$

Average: $\sim 1.29\text{e-}4$

Average, similar airlines: $\sim 1.5\text{e-}4$

Again, United is employing far more ground crew than all of its competitors, with no discernible difference in delay time

Recommendation: United should slim down its ground crews



Case Study - Flights per Month

United: ~51,977 w/ 8,501,631 pass. (163.6 per)

Southwest: ~111,171 w/ 13,382,999 pass. (120.4 per)

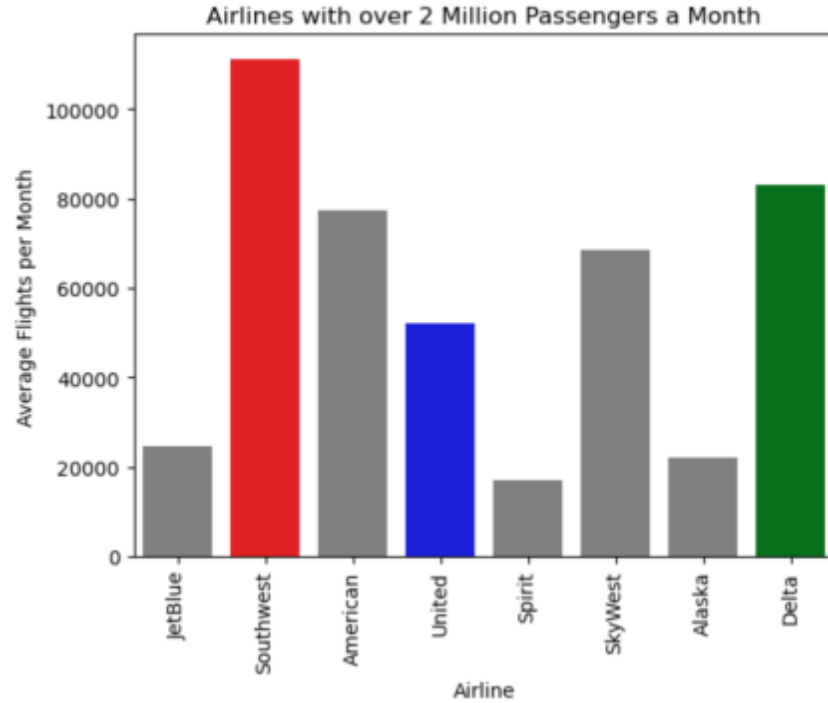
Delta: ~83,069 w/ 12,460,183 pass. (150.0 per)

Average: ~35,906 w/ 4,070,878 pass. (113.4 per)

Average, similar airlines: ~56,936 w/ 7,290,721 pass. (128.1 per)

United is operating far fewer flights than its competitors, operating with 13.6 more people per flight than delta, and 35.5 more people than similar airlines

Recommendation: United needs to start decreasing the number of people per flight by increasing the number of flights offered, or changing plane size

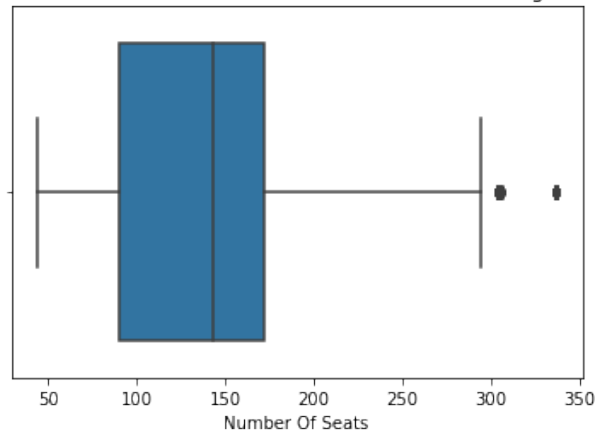


Airplanes

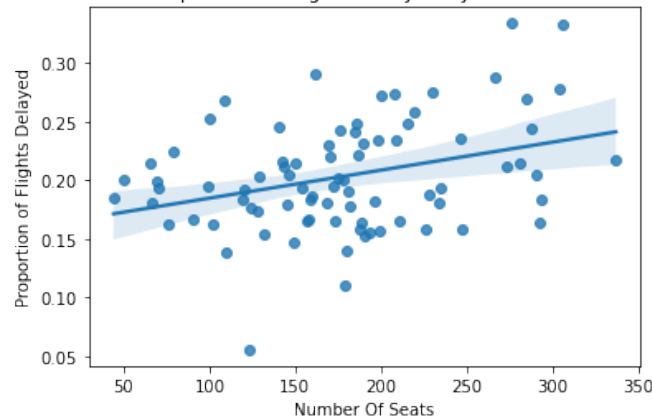
Exploratory Data Analysis

Does the size of the airplane affect how likely it is to be delayed?

Distribution Of The Number Of Seats Across All Flights



Proportion Of Flights Delayed By Plane Size



Airplanes

Categorizing Plane Size

Small

$43 < \text{Number Of Seats} \leq 124$

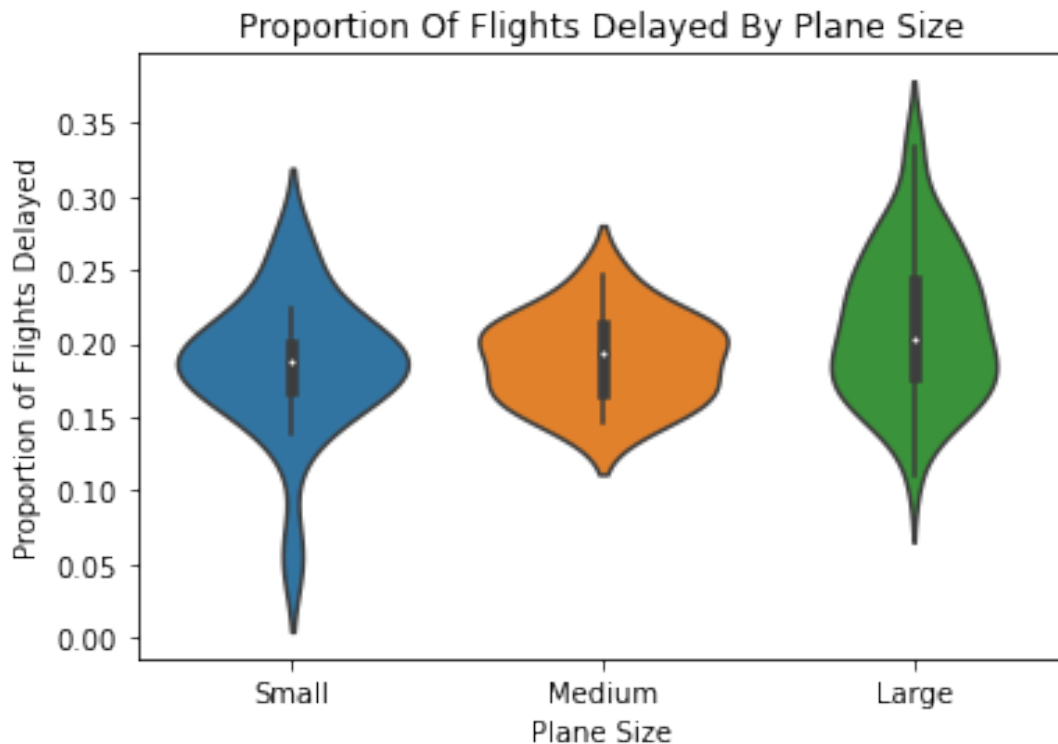
Medium

$124 < \text{Number Of Seats} \leq 158$

Large

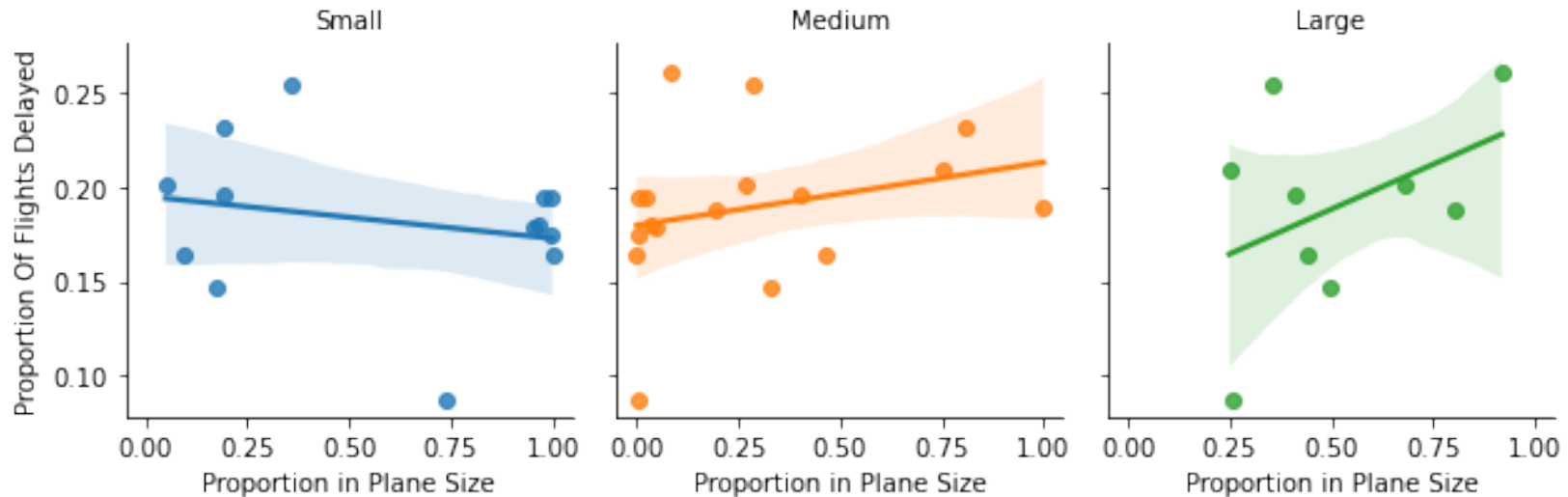
$158 < \text{Number Of Seats} \leq 337$

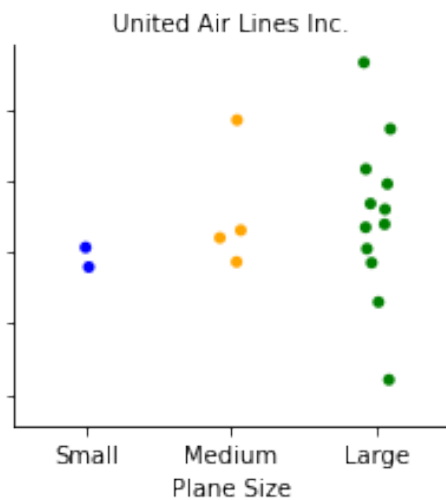
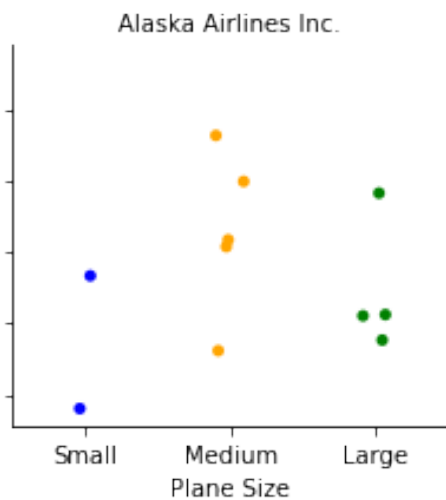
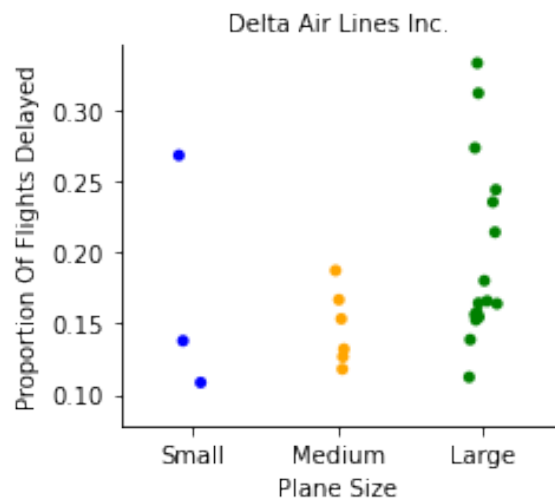
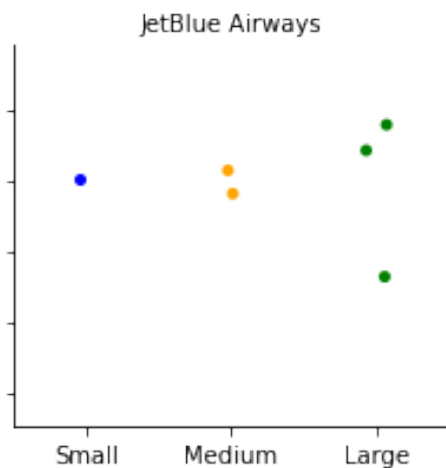
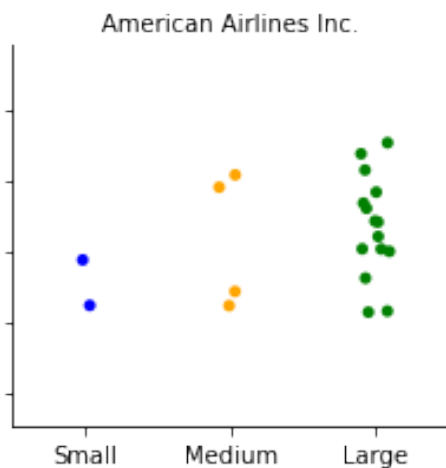
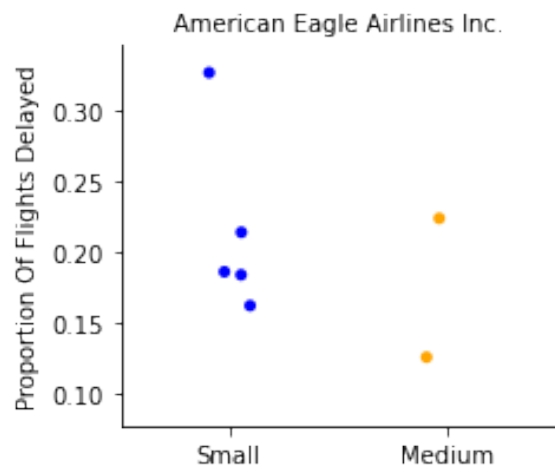
~ 2.16 Million Flights in Each Category



Varying By Carrier

Different plane layouts and boarding procedures likely lead to differences in performance for the same plane size





Plane Size Options

Does having more plane sizes lead to less delays overall?

Delta Air Lines Inc.	25
American Airlines Inc.	21
United Air Lines Inc.	18
Alaska Airlines Inc.	11
American Eagle Airlines Inc.	7
JetBlue Airways	6
SkyWest Airlines Inc.	6
Mesa Airlines Inc.	6
Midwest Airline, Inc.	5
Endeavor Air Inc.	5
Comair Inc.	5
Hawaiian Airlines Inc.	5
Spirit Air Lines	5
Atlantic Southeast Airlines	4
Southwest Airlines Co.	4
Frontier Airlines Inc.	4
Allegiant Air	2

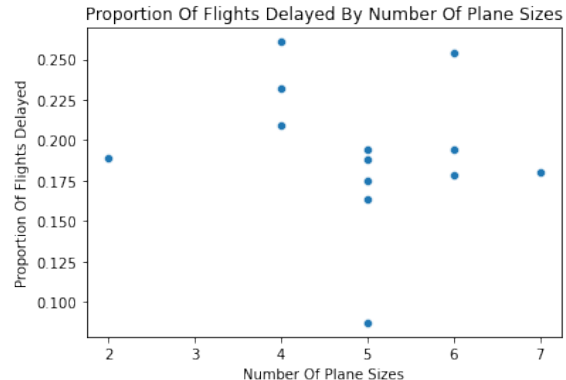
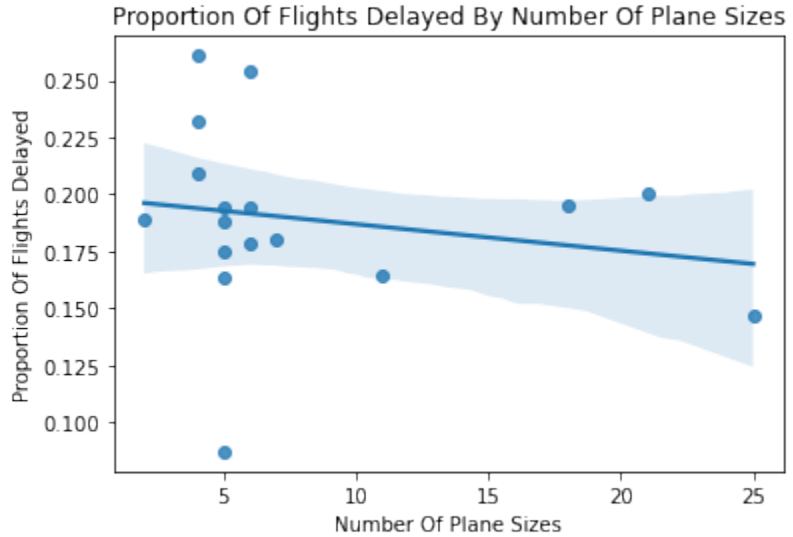
Plane Size Options

Does having more plane sizes lead to less delays overall?

Correlation is very low

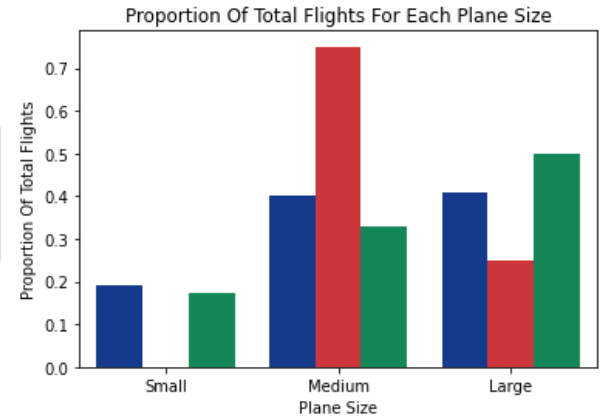
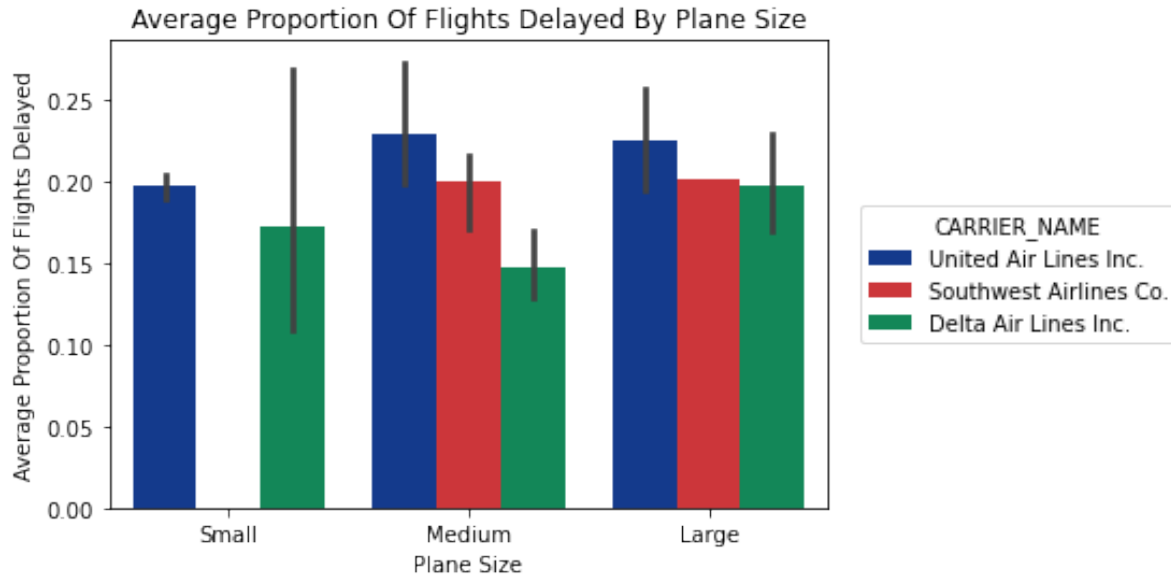
$$r^2 = 0.037269$$

No significant trend when outliers removed



Focusing On United

What airplanes should United Airlines purchase to decrease future delays?



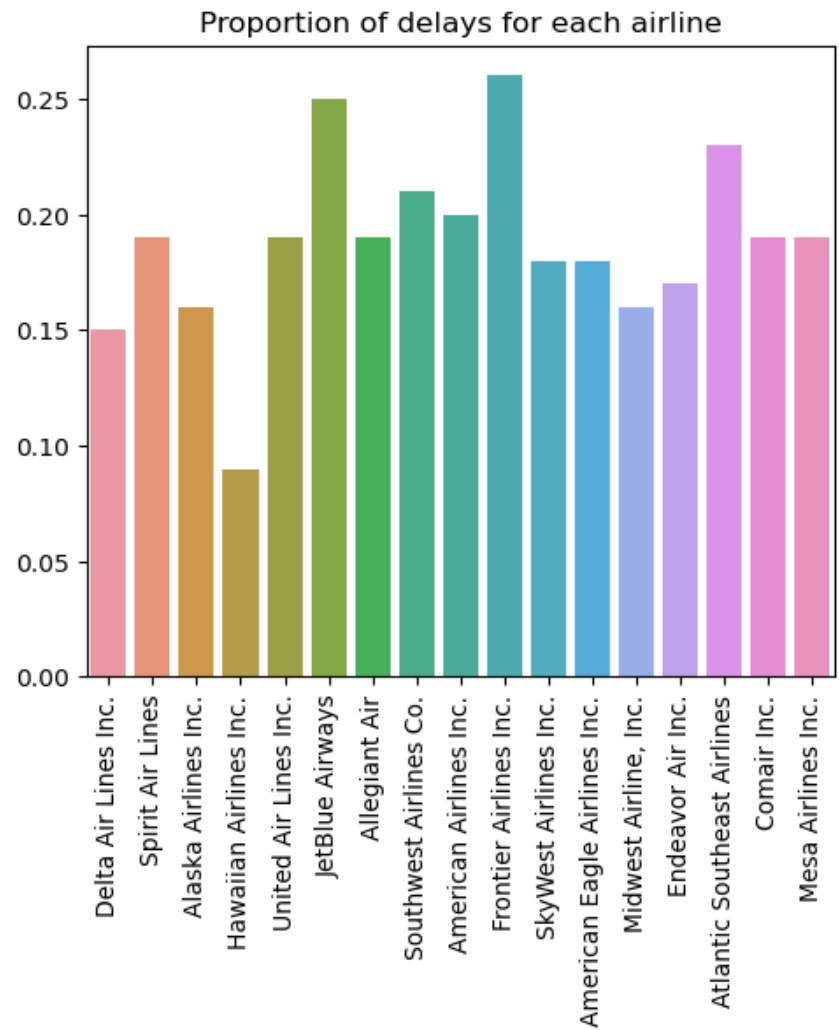
4th EDA: United and weather

is the way United operates in different weather conditions responsible for a bigger probability of delay?

- 1) United is the 6th most likely airline to be delayed. How much of that is due to “bad” weather?
- 2) Compared to other airlines, is it the case that United flies more in bad weather?
- 3) If yes, what factors of “bad” weather are most likely to affect its
- 4) Case study of airline of biggest probability of delays and airline of lowest prob of delay : how does their weather indicator look like

Proportion of delays for airlines

- Biggest probability of being delayed the most: Frontier Airlines
- Lowest probability of being delayed: Hawaiian Airlines Inc



Creating the weather indicator

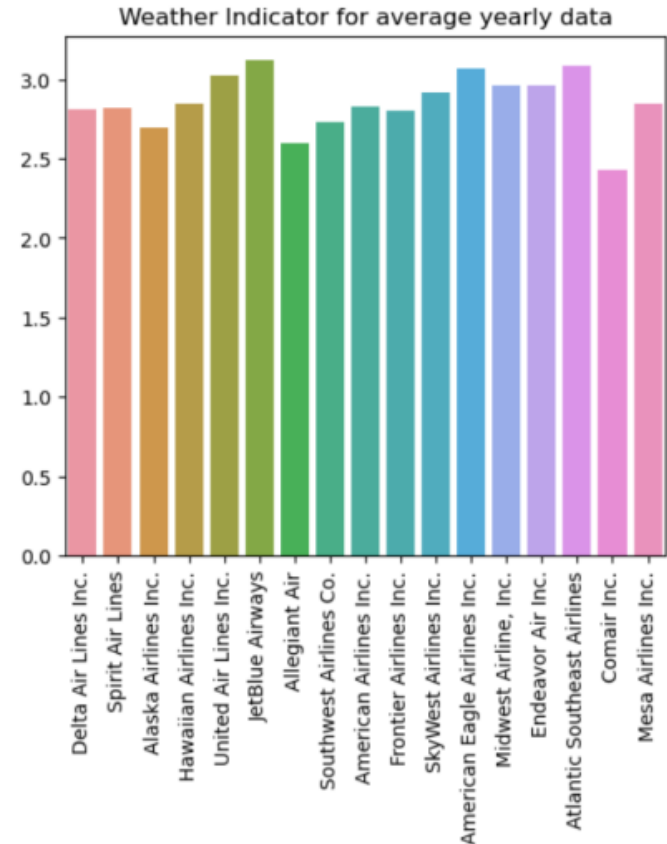
Weather indicator = (snow + rain + wind) / 3

United = **4th position**

- Flying more frequently so more flights happen in bad weather - **9%** of all flights in dataset are for United

Notice: Comair Inc <**United**< JetBlue Airways

Conclusion 1: United has a high weather indicator which could be causing the delays



Does weather cause delays?

- Split data into subsets to notice monthly trends
 - Calculate delays per month and indicator per month
- Find correlation between delays and weather indicator

Weather indicator only:

Comair Inc <United< JetBlue Airways

Weather indicator + correlation:

JetBlue Airways <United< Comair Inc

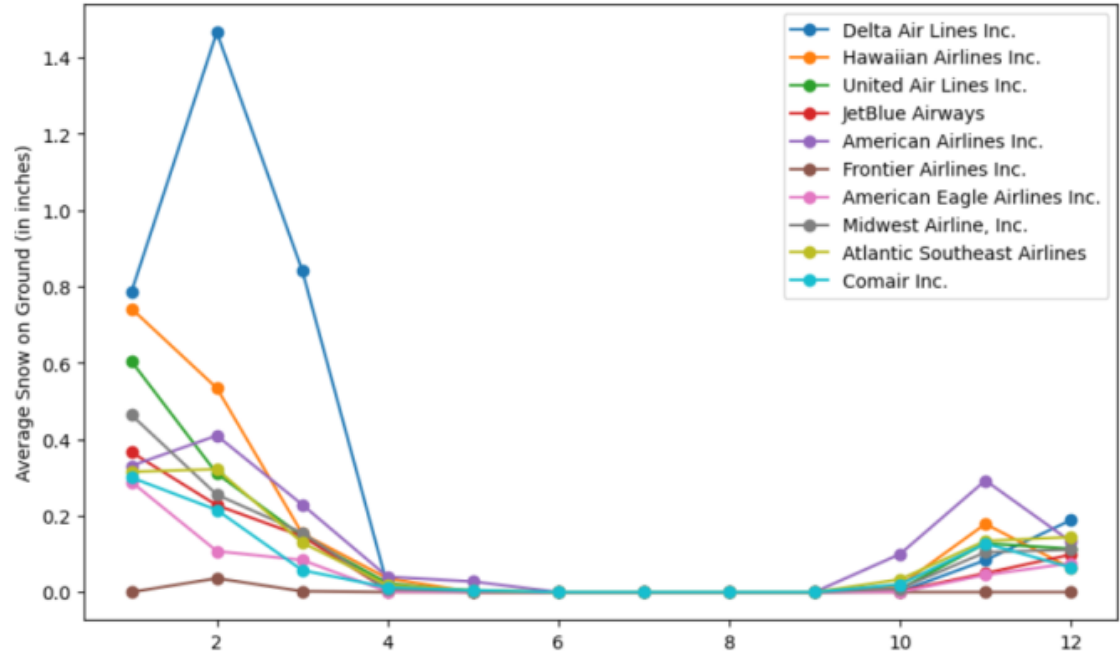
Conclusion 2: United's delays are likely not to be due to bad weather

Endeavor Air Inc.	0.401403
Atlantic Southeast Airlines	0.396127
Comair Inc.	-0.324852
American Airlines Inc.	-0.300407
United Air Lines Inc.	-0.165776
Allegiant Air	-0.139086
Southwest Airlines Co.	0.134725
American Eagle Airlines Inc.	-0.125721
JetBlue Airways	-0.124465
SkyWest Airlines Inc.	-0.124411
Delta Air Lines Inc.	0.121742
Spirit Air Lines	-0.075680
Mesa Airlines Inc.	-0.071582
Frontier Airlines Inc.	-0.070815
Alaska Airlines Inc.	0.046326
Hawaiian Airlines Inc.	0.030387
Midwest Airline, Inc.	0.009300
dtype:	float64

Snow: Potential for weather recommendations?

No, United flying in an average amount of **snow**

Comair Inc =United=JetBlue Airways

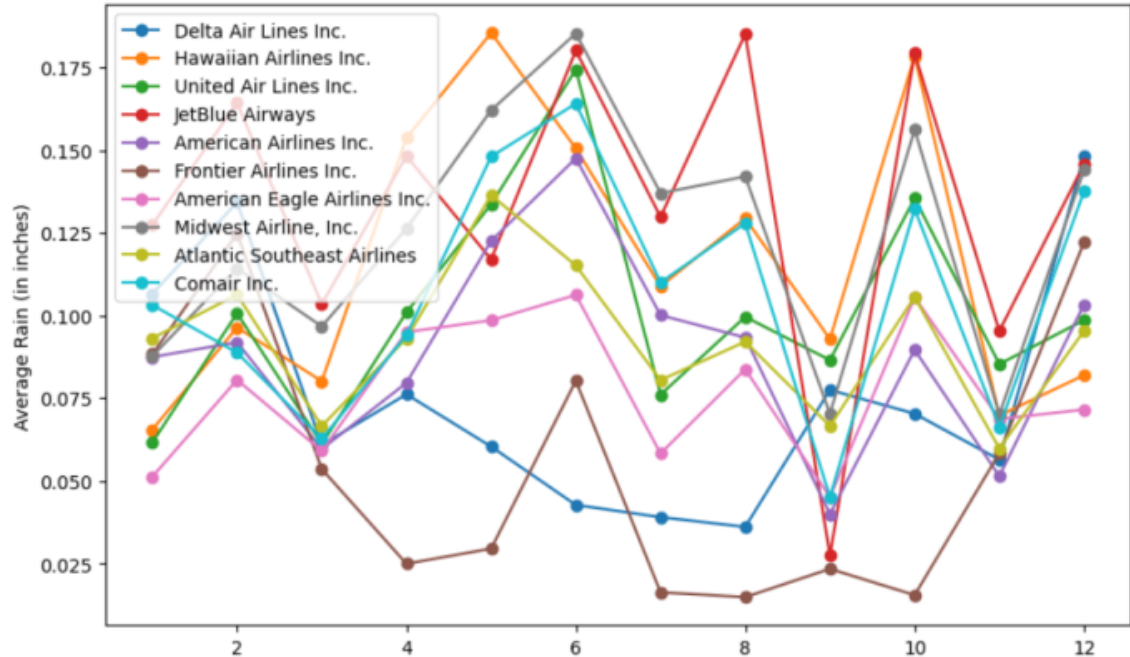


Rain: Potential for weather recommendations?

United vs other airlines:

Yes, rain might be a weather factor that is weighing more heavily or the weather indicator

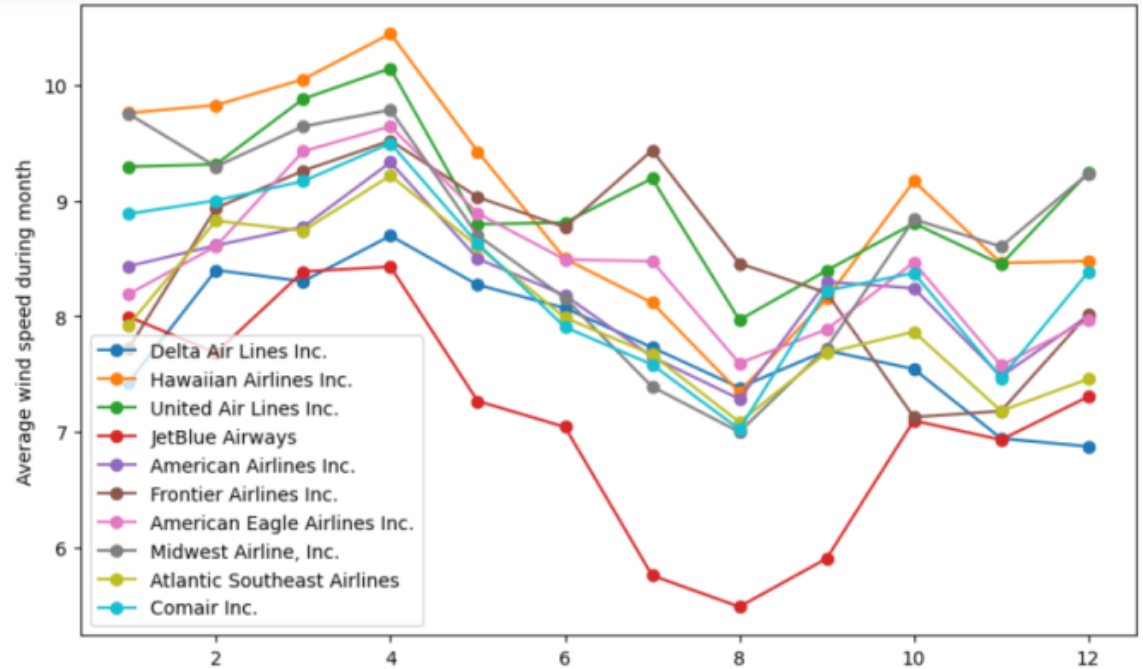
Comair Inc =United<<JetBlue Airways



Wind: Potential for weather recommendations?

Yes, United is flying in highest wind speed compared to other airlines

JetBlue Airways < Comair Inc < United



Summary

1. Closest Neighbor Approach
 - a. Select airports with a lower probability of delay
2. Emulate the Winners, Avoid the Losers
 - a. Fly More
 - b. Less Employees
3. Weather the Storm or Don't
 - a. Delays not due to weather - could do better with handling rain, potentially snow
4. Bigger Bang for your Buck
 - a. Purchase Smaller Planes

